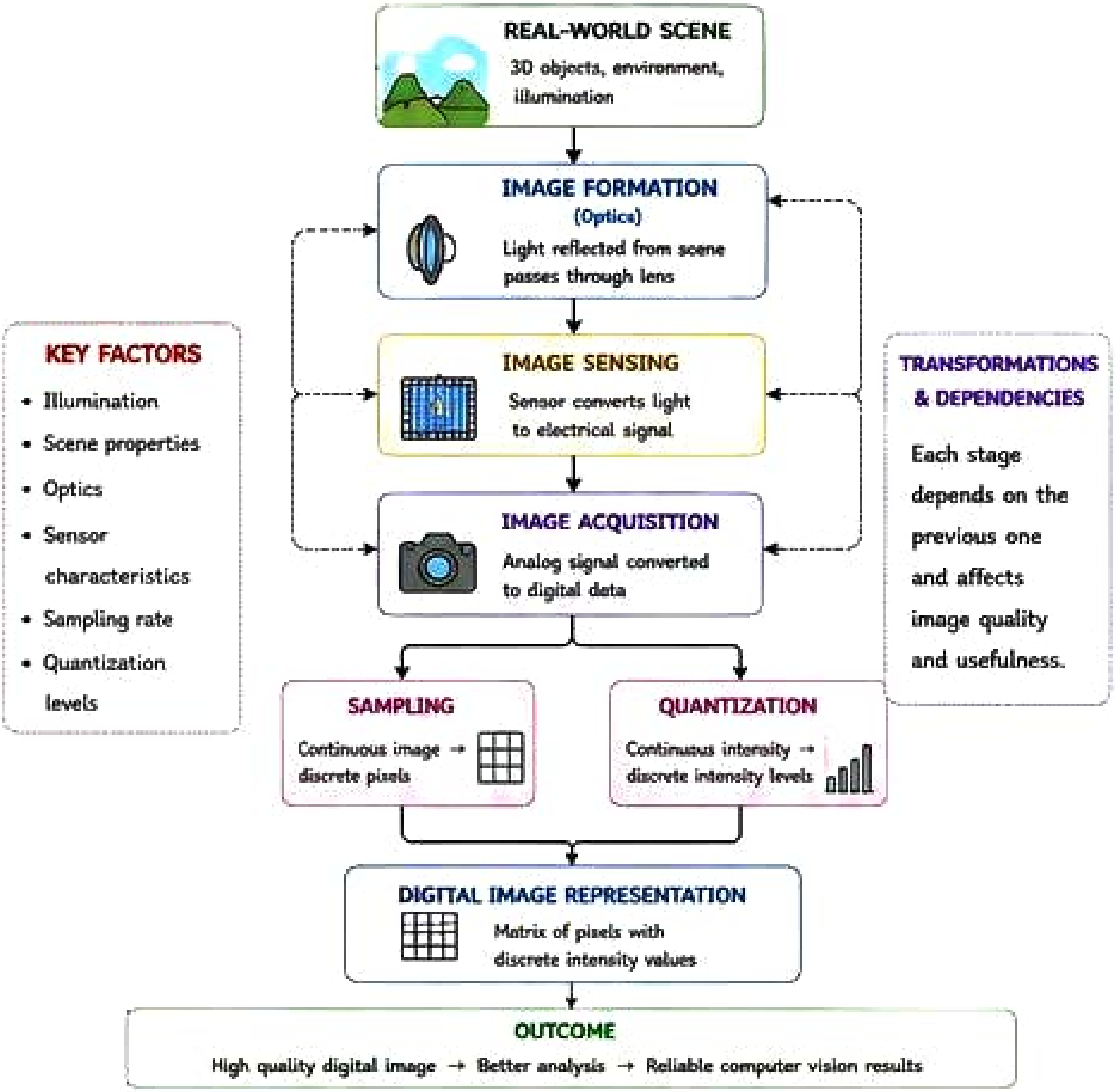
QUESTION – 1

From Real-World Scene to Digital Image Representation – Complete Pipeline

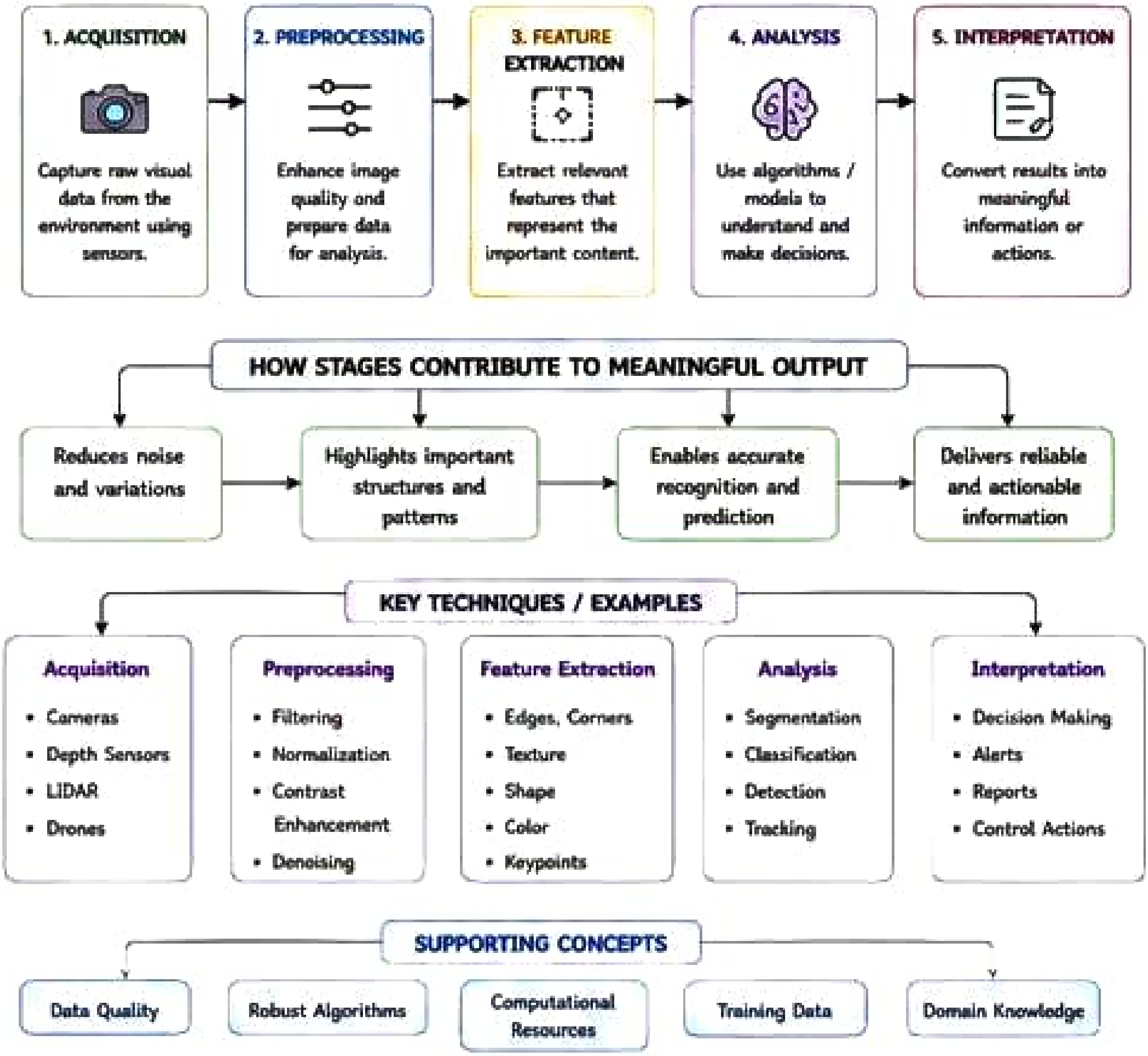


Recent Literature (Examples)

1. Gonzalez, R. C., & Woods, R. E. (2018). Digital Image Processing (4th Ed.). Pearson.
2. Szeliski, R. (2022). Computer Vision: Algorithms and Applications (2nd Ed.). Springer.
3. Jain, A. K. et al. (2000). Fundamentals of Digital Image Processing. Prentice Hall.
4. Brownlee, J. (2019). Deep Learning for Computer Vision. Machine Learning Mastery.
5. Sonka, M., Hlavac, V., & Boyle, R. (2014). Image Processing, Analysis, and Machine Vision. Cengage.

QUESTION – 2

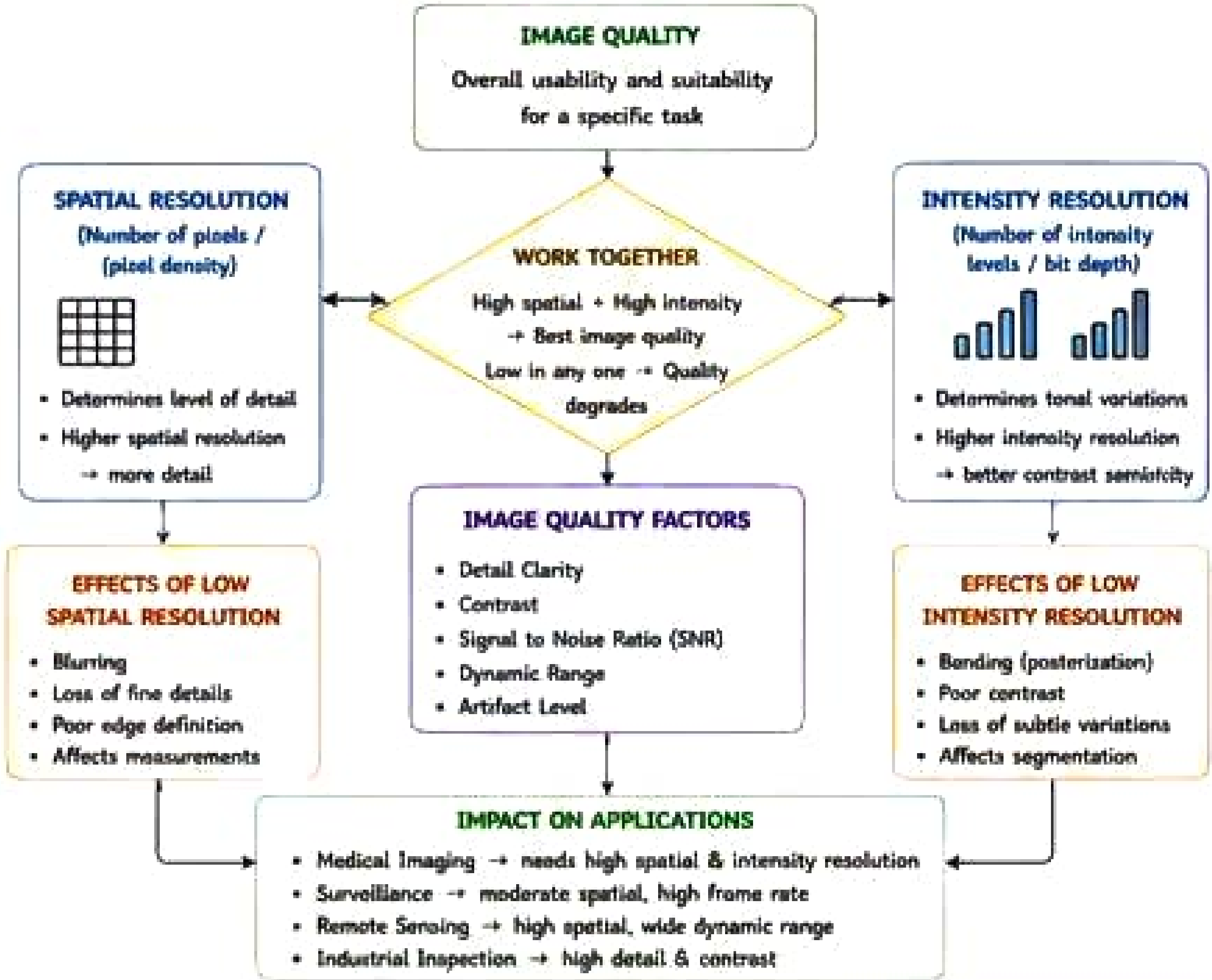
Visual Data Handling Pipeline in a Computer Vision System



Recent Literature (Examples)

1. Shapiro, L. G., & Stockman, G. C. (2021). Computer Vision. Pearson.
2. Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.
3. Ludovicos, A. et al. (2018). Deep Learning for Computer Vision: A Review.
4. Duda, R. O., Hart, P. E., & Stork, D. G. (2019). Pattern Classification (2nd Ed.). Wiley.
5. Zhou, Z.-H. (2021). Machine Learning for Computer Vision. Springer.

QUESTION – 3 Image Resolution and Image Quality – Relationships and Impact

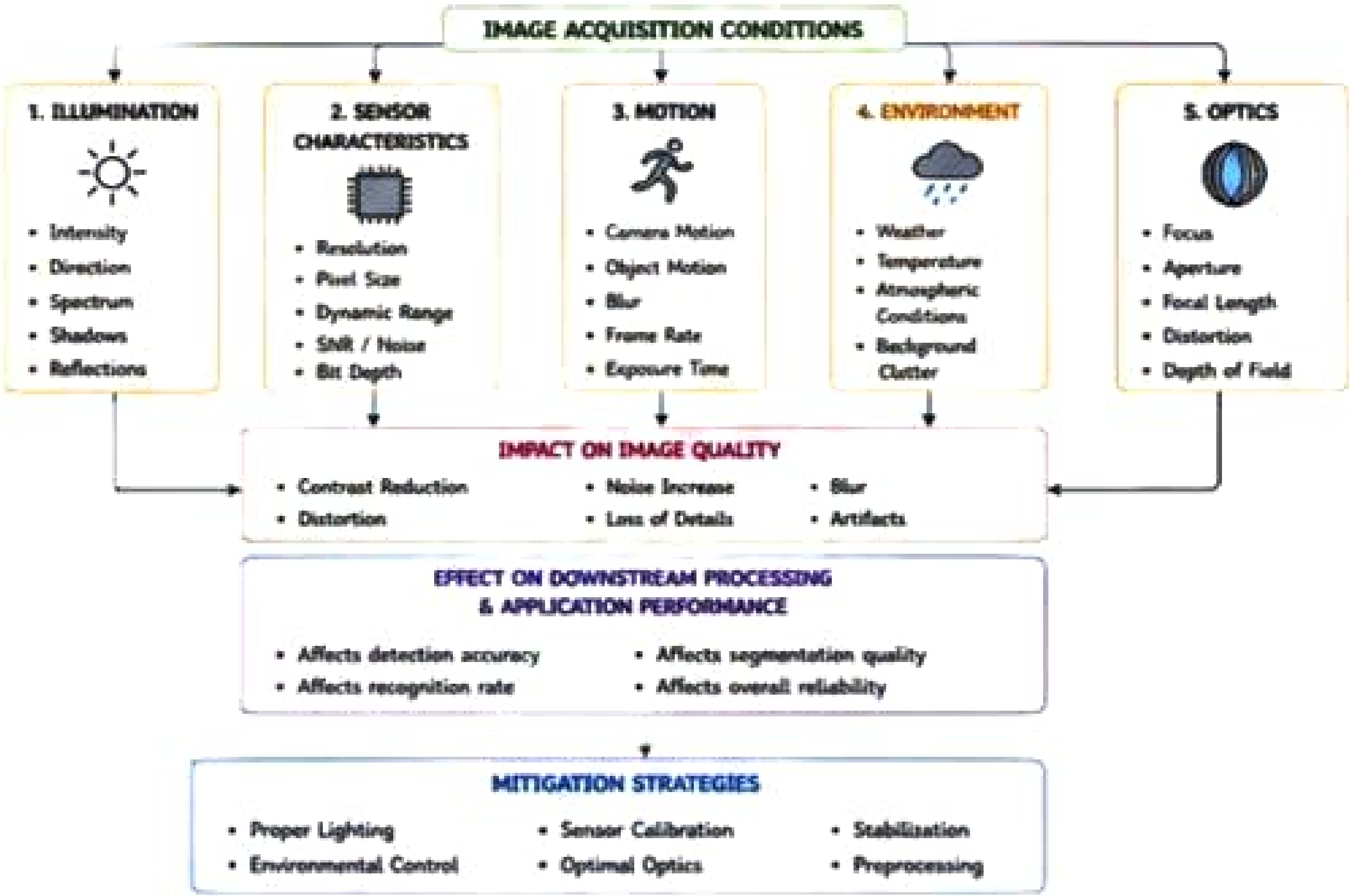


Recent Literature (Examples)

1. Gonzalez, R. C., & Woods, R. E. (2018). Digital Image Processing (4th Ed.). Pearson.
2. Pratt, W. K. (2010). Digital Image Processing (5th Ed.). Wiley.
3. Pizer, S. M. (2001). Introduction to Digital Image Processing. CRC Press.
4. Mohan, A. et al. (2020). A Review on Image Quality Assessment. IET Image Processing.
5. Wang, Z., Bovik, A. C., Sheikh, H. R., & Simoncelli, E. P. (2019). Image Quality Assessment: From Error Visibility to Structural Similarity. IEEE T-IP.

QUESTION – 4

Factors Influencing Image Acquisition Conditions and Their Impact



Recent Literature (Examples)

1. Holst, G. C. (2012). CCD Arrays, Cameras, and Displays. SPIE Press.
2. Jain, A. K. (2019). Fundamatiorls of Digital Image Processing. Pearson.
3. Xu, Y., et al. (2020). A Review on Image Acquisition in Computer Vision. Sensors.
4. Li, X., & Lee, G. G. (2018). Motion Blur Analysis for Image Acquisition Systems. IEEE T-IM.
5. Fairchild, M. D. (2013). Color Appearance Models (3rd Ed.). Wiley.

QUESTION – 5

COMPUTER VISION SYSTEM CAPABILITIES AND APPROPRIATE APPLICATIONS

COMPUTER VISION

Enables machines to perceive, interpret and understand visual information from the real world.

DEPENDS ON KEY CAPABILITIES

1. IMAGE ACQUISITION
- Capture images from the real world
- Handles different sensors and conditions

2. IMAGE ENHANCEMENT
- Improve quality and clarity
- Reduce noise, enhance contrast, adjust illumination

3. FEATURE EXTRACTION
- Detect edges, textures, shapes, corners
- Extract key features and descriptors

4. OBJECT RECOGNITION
- Identify and classify objects, patterns, symbols
- Use of ML/DL models

5. TRACKING & MOTION ANALYSIS
- Track objects across frames
- Estimate motion, speed and trajectories

6. SCENE UNDERSTANDING
- Understand context, and relationships between objects
- 2D / 3D scene interpretation

7. DECISION & ACTION
- Make intelligent decisions
- Trigger actions, alerts or control systems

INFLUENCE ON APPLICATION EFFECTIVENESS

- Higher accuracy and reliability
- Better performance and efficiency
- Ability to handle complex real-world conditions
- Enables automation and real-time decision making

ENABLES A WIDE RANGE OF VISION-BASED APPLICATIONS

HEALTHCARE
- Medical image analysis
- Disease detection and diagnosis

AUTONOMOUS VEHICLES
- Lane detection
- Obstacle detection
- Navigation & path planning

INDUSTRIAL AUTOMATION
- Defect detection
- Quality control
- Process automation

SURVEILLANCE & SECURITY
- Face recognition
- Intrusion detection
- Anomaly detection
- Monitoring

AGRICULTURE
- Crop monitoring
- Pest detection
- Yield estimation
- Precision farming

RETAIL & ANALYTICS
- Customer behavior analysis
- People counting
- Inventory tracking

ROBOTICS
- Object manipulation
- Environment perception
- Autonomous tasks

RELATIONSHIP SUMMARY

System Capabilities (What the system can do) → Application Suitability (What the system is best suited for) → Performance Impact (Accuracy, speed, reliability) → Real-World Value (Better decisions, automation, safety)

RECENT LITERATURE (EXAMPLES)

1. Szeliski, R. (2022). Computer Vision: Algorithms and Applications (2nd Ed.). Springer.
2. Chen, Y., et al. (2020). Applications of Computer Vision: A Survey. Artificial Intelligence Review.
3. Ma, W., & Liu, Z. (2019). Vision-Based Applications in Industry: A Review. IEEE Access.
4. Litjens, G., et al. (2017). A Survey on Deep Learning in Medical Image Analysis. Medical Image Analysis.
5. Hosseini, M. S., et al. (2021). Deep Learning for Vision-Based Applications: A Review. Computers & Electrical Engineering.